

Pythagoras, mensuration and 3D - Set 2 - Solutions

Specification M5.7, M5.11-M5.13, M5.15-M5.17.

Answer key

Question	Answer	Topic
1	C	Pythagoras, mensuration and 3D
2	D	Pythagoras, mensuration and 3D
3	D	Pythagoras, mensuration and 3D
4	D	Pythagoras, mensuration and 3D
5	C	Pythagoras, mensuration and 3D

Worked solutions

Question 1 MEN-16 · Pythagoras, mensuration and 3D · spec M5.18

From the top of a cliff 40 m high, the angle of depression of a boat is 30° . How far is the boat from the foot of the cliff?

Answer: C

$$\tan 30^\circ = \frac{40}{d} \Rightarrow d = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}.$$

Question 2 X2-29 · Pythagoras, mensuration and 3D · spec M5.16

P: 'a solid cylinder and a solid cube have equal volumes'. *Q*: 'the cylinder and the cube have equal surface areas'. Which one of the following is true?

Answer: D

Equal volumes say nothing about surface areas (a long thin cylinder has a large surface area), and equal surface areas say nothing about volumes. Neither implies the other.

Question 3 MEN-02 · Pythagoras, mensuration and 3D · spec M5.16

A cylinder has radius r and height h . Its volume is 54π and $h = 2r$. Find its total surface area.

Answer: D

$$\pi r^2 \cdot 2r = 2\pi r^3 = 54\pi \Rightarrow r = 3, h = 6. \text{ Surface area} = 2\pi r^2 + 2\pi rh = 18\pi + 36\pi = 54\pi.$$

Question 4 MEN-10 · Pythagoras, mensuration and 3D · spec M5.7

A square-based pyramid has base side 6 and all slant edges of length 9. Find its height.

Answer: D

$$\text{Half the base diagonal is } 3\sqrt{2}, \text{ so } h^2 = 81 - 18 = 63.$$

Question 5 MEN-13 · Pythagoras, mensuration and 3D · spec M5.7

The points $A(1, 2)$, $B(4, 6)$ and $C(k, 2)$ with $k > 1$ form a triangle with a right angle at B . Find k .

Answer: C

$$\text{Gradient } AB = \frac{4}{3}, \text{ so } BC \text{ must have gradient } -\frac{3}{4}: \frac{2-6}{k-4} = -\frac{3}{4} \Rightarrow 16 = 3(k-4) \Rightarrow k = \frac{28}{3}.$$